

A proof of Conjecture 00000007630

Clifford algebras of signatures $(0, 1, 0)$ and $(0, 0, 1)$: same idempotents, different zero divisors

Jackmeson1

1 The conjecture

Conjecture 00000007630 reads (English version; the Chinese version agrees, and it says “explicit signature pair” for the last clause):

Definition: Idempotent spectra and zero-divisor spectra are two layers. *Conjecture:* There exist two Clifford algebras with the same idempotent spectrum but different zero-divisor spectra, and the separation is realized by an explicit signature pair with the same idempotents but different zero divisors.

This is an existence claim. We exhibit an explicit pair of real quadratic forms with signatures $(0, 1, 0)$ and $(0, 0, 1)$. We prove that their Clifford algebras have the same idempotents and different zero divisors.

2 Reading and definitions

Definition 1. For a real quadratic form Q on V , the Clifford algebra $\text{Cl}(Q)$ is the quotient of the tensor algebra $T(V)$ by the relations $v \otimes v = Q(v) \cdot 1$. In Lean this is Mathlib’s `CliffordAlgebra` $\mathbb{Q}$, with generator map $\iota: V \rightarrow \text{Cl}(Q)$. Degenerate forms are allowed, as in Mathlib; the Clifford algebra of the zero form is the exterior algebra. The *signature* of Q is its Sylvester signature (p, q, r) : the numbers of positive, negative and null directions.

Definition 2. For an algebra A , the idempotent spectrum is $\text{Idem}(A) = \{x : x^2 = x\}$. The zero-divisor spectrum is $\text{ZD}(A) = \{x : \exists y \neq 0, xy = 0 \text{ or } yx = 0\}$. With this convention $0 \in \text{ZD}(A)$; nonzero zero divisors are reported separately.

The explicit signature pair. On $V = \mathbb{R}$:

- $Q_1(t) = -t^2$ (Mathlib’s `CliffordAlgebraComplex.Q`). It is negative definite, so its signature is $(0, 1, 0)$. Mathlib’s `CliffordAlgebraComplex.equiv` gives $\text{Cl}(Q_1) \cong \mathbb{C}$.
- $Q_0(t) = 0$. It is totally null, so its signature is $(0, 0, 1)$. Mathlib’s `CliffordAlgebraDualNumber.equiv` gives $\text{Cl}(Q_0) \cong \mathbb{R}[\varepsilon]$ (the dual numbers, with $\varepsilon^2 = 0$).

3 The proof

Lemma 3. In the dual numbers $\mathbb{R}[\varepsilon]$, the only idempotents are 0 and 1.

Proof. Write $z = a + b\varepsilon$. Then $z^2 = a^2 + 2ab\varepsilon$, so $z^2 = z$ gives $a^2 = a$ and $2ab = b$. If $a = 0$ then $b = 0$; if $a = 1$ then $2b = b$, so $b = 0$. $\square$

Theorem 4 (same idempotents). $\text{Idem}(\text{Cl}(Q_1)) = \{0, 1\}$ and $\text{Idem}(\text{Cl}(Q_0)) = \{0, 1\}$. In each algebra $0 \neq 1$, so each has exactly two idempotents.

Proof. $\text{Cl}(Q_1) \cong \mathbb{C}$ has no zero divisors, and in such a ring $x^2 = x$ forces $x \in \{0, 1\}$. For $\text{Cl}(Q_0)$, transport Lemma 3 along the algebra isomorphism. $\square$

Theorem 5 (different zero divisors). $\text{ZD}(\text{Cl}(Q_1)) = \{0\}$. In contrast, $\text{ZD}(\text{Cl}(Q_0)) = \iota(\mathbb{R}) = \{\iota(r) : r \in \mathbb{R}\}$, and $e = \iota(1)$ satisfies $e \neq 0$ and $e^2 = 0$. Consequently $\text{Cl}(Q_1) \not\cong \text{Cl}(Q_0)$ as $\mathbb{R}$ -algebras.

Proof. The first claim holds because $\mathbb{C}$ has no zero divisors. In $\text{Cl}(Q_0)$ we have $\iota(r)\iota(s) = rs\iota(1)^2 = rsQ_0(1) = 0$, and $\iota(1) \mapsto \varepsilon \neq 0$. Conversely, if the image $a + b\varepsilon$ of x has $a \neq 0$, then it is a unit, so x is not a zero divisor. If $a = 0$, then $x = \iota(b)$. An algebra isomorphism would carry the nonzero zero divisor e to a nonzero zero divisor of $\text{Cl}(Q_1)$, which does not exist. $\square$

Theorems 4 and 5 give the conjecture. $\square$

Remark 6 (scope and conventions). (1) “The same idempotent spectrum” is proved in three senses: equal idempotent sets $\{0, 1\}$, equal cardinality (2), and, implicitly, the obvious bijection $0 \mapsto 0$, $1 \mapsto 1$. (2) “Different zero-divisor spectra” holds whether or not 0 counts as a zero divisor: $\text{Cl}(Q_1)$ has no nonzero zero divisor, while $\text{Cl}(Q_0)$ has a whole line of them. (3) The second form is degenerate. Among *nondegenerate* real Clifford algebras, “only the trivial idempotents” already forces a division algebra $(\mathbb{R}, \mathbb{C}, \mathbb{H})$, since such algebras are semisimple (Wedderburn theory). This standard fact is context only and is not used in the proof. So under the reading “idempotent set = $\{0, 1\}$ ”, a separating pair must involve a degenerate signature. This is why the null direction $r = 1$ appears.

4 Lean formalization

File `lean/Conjecture7630/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1). It defines `idempotents`, `zeroDivisors`, `Q1 := CliffordAlgebraComplex.Q`, `Q0 := 0`, `Cl1`, `Cl0`, and `gen :=  $\iota(1)$` . The theorems are:

- `Q1_negDef`, `Q0_null`: the signatures.
- `dual_idem`, `idempotents_Cl1`, `idempotents_Cl0`: Lemma 3 and Theorem 4.
- `zeroDivisors_Cl1`, `zeroDivisors_Cl0`, `gen_ne_zero`, `gen_mul_gen`, `not_isomorphic`: Theorem 5.
- `separation`: the main theorem. It bundles both signatures, both idempotent sets = $\{0, 1\}$ with `ncard 2`, both zero-divisor sets, the nonzero square-zero element, and non-isomorphism.

lake build succeeds. `#print axioms` for `separation`, `not_isomorphic` and `zeroDivisors_Cl0` reports only `[propext, Classical.choice, Quot.sound]`. The code contains no `sorry` and no `native_decide`.